Project 1: 2026

Scenario:

The Best Buys Supermarket tracks sales for seven departments using arrays. The program should:

1. Store department codes in a one-dimensional array. Each department has a unique three-character code.
2. Store sales transactions in a two-dimensional array, where:
    - There are seven rows, one for each department.
    - Each department can store up to 1,000 sales transactions.
3. Initialize all sales transactions to zero at the start of the program.
4. Allow the user to enter sales transactions by inputting a department code and a sales amount.
5. Find the department index using the department code and insert the sales amount into the next available slot.
6. Prevent errors, such as:
    - Invalid department codes.
    - More than 1,000 sales transactions per department.
    - Negative or invalid sales amounts.
7. At the end of the program, display:
    - Each department's code.
    - The total sales amount for that department.

1.1 What is the difference between a one-dimensional and a two-dimensional array?

- Explain the structure and purpose of **1D arrays** and **2D arrays** with examples.
- Discuss when and why a **2D array** is preferable for storing sales transactions in this scenario.

1.2 Construct an IPO (Input-Processing-Output) table to illustrate the program's functionality.

- Break down how the system works in terms of **inputs, processes, and outputs**.

1.3 Explain the reasoning behind your IPO design.

- Justify the approach taken to implementing the program.
- Discuss why using **a 1D array for department codes, a 2D array for sales, and an index tracking system** is an optimal solution.

1.4 What validation measures have been implemented in this program?

- Explain how the program ensures:
    - Only valid department codes are used.
    - Sales amounts are correctly formatted and within acceptable limits.
    - No department exceeds 1,000 transactions.

1.5 How would you improve or extend this program?

- Suggest potential enhancements in order to improve the efficiency of the program.

Mark Allocation

Criteria	Mark
Understanding of Arrays (1D vs 2D)	5
2. IPO Diagram (Input, Process, Output clarity)	10
3. Explanation of IPO Design & Program Structure	10
4. Code Implementation (Correctness, Efficiency, Readability)	15
5. Error Handling & Validation Techniques	10
6. Program Output Accuracy (Correct totals for each department)	10
7. Additional Features & Improvements	5
Presentation	5
Total	70 Marks

IMPORTANT INFORMATION for Project 1

- This is a group assignment. You will complete this assignment in groups of between 3 and 4 learners. **If you work on your own your mark will be capped at 50%**

- There will only be a single submission for each group. Further communication in this regard will be sent out during the week.
- Belgium Campus makes use of software that can **scan for plagiarism** and a student caught doing this will get 0 for this assignment.
- Late assignments will not be accepted and missing the deadline is an automatic 0.
- All work should be submitted as a PDF Document.